



Professional Educational Materials

Student Number

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2023

HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks: 100

Section I – 10 marks (pages 2-6)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7-35)

- Attempt Questions 11-35
- Allow about 2 hours and 45 minutes for this section

Directions to School or College

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Section I

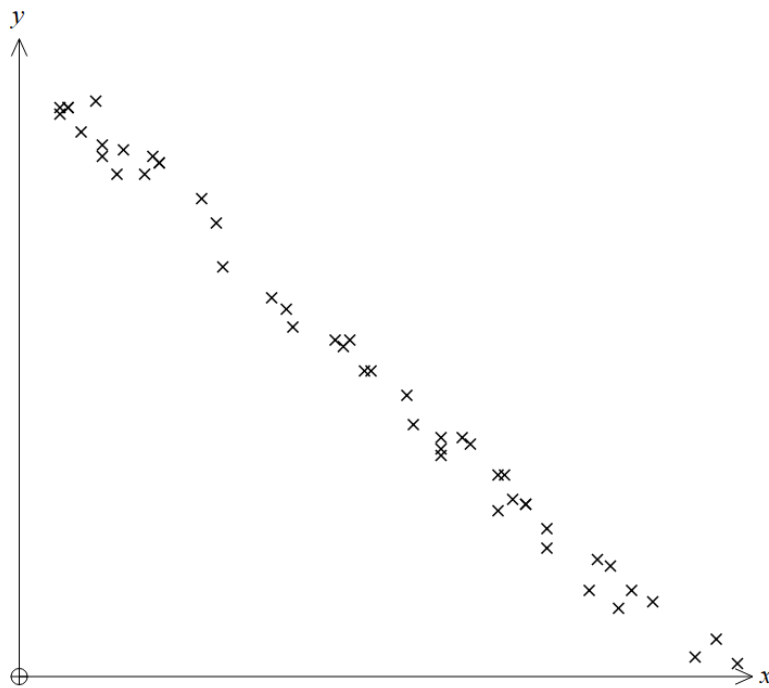
10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. The scatter plot below shows the relationship between x and y . Which of the following may represent the relationship between x and y ?



- A. x remains unchanged when y increases.
- B. y varies inversely as x .
- C. y increases when x increases.
- D. y decreases when x increases.

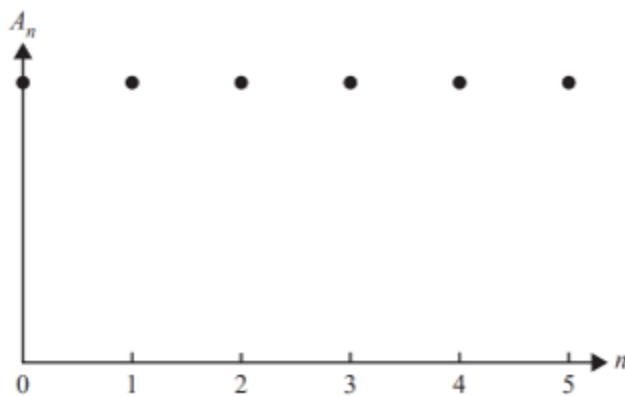
2. Simplify $\sqrt{x^2 + 2x + 1} - \sqrt{x^2 - 2x + 1}$.

- A. -2
- B. 2
- C. $2x$
- D. $2x + 2$

3. What is the value of $7^{334} \left(\frac{-1}{343} \right)^{111}$?

- A. -7
- B. $-\frac{1}{7}$
- C. 1
- D. 7

4. The graph below represents the value A_n , in dollars, of an annuity investment for five time periods.



A recurrence relation that could match this graphical representation is.

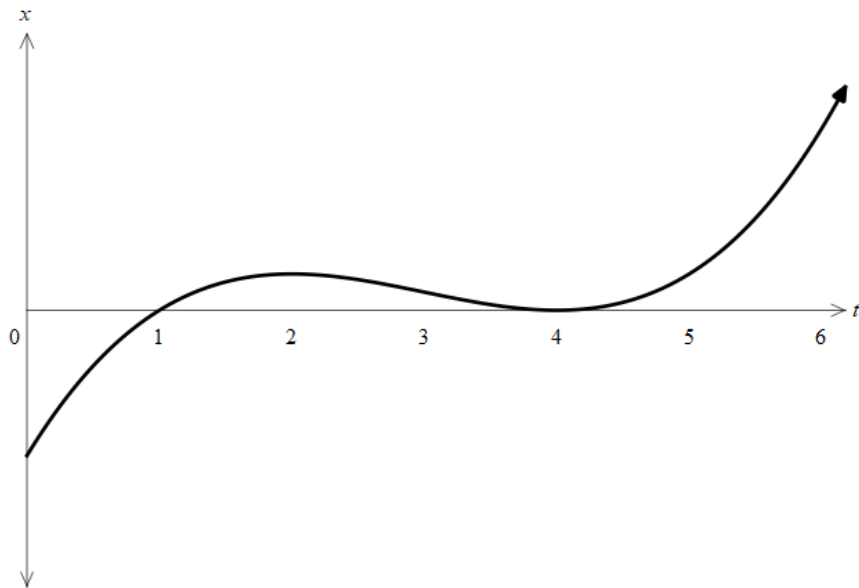
- A. $A_0 = 200000$, $A_{n+1} = 1.015A_n - 2500$
- B. $A_0 = 200000$, $A_{n+1} = 1.025A_n - 5000$
- C. $A_0 = 200000$, $A_{n+1} = 1.035A_n - 5500$
- D. $A_0 = 200000$, $A_{n+1} = 1.04A_n - 6000$

5. The z-score of Tom in a Science examination is -2. Tom scores 34 in Science, whilst the mean of scores is 50.

What is the standard deviation of scores in the Science examination?

- A. 4 marks
- B. 8 marks
- C. 12 marks
- D. 16 marks

6. The displacement, x metres, from the origin of a particle moving in a straight line at any time, t seconds, is shown in the graph below.



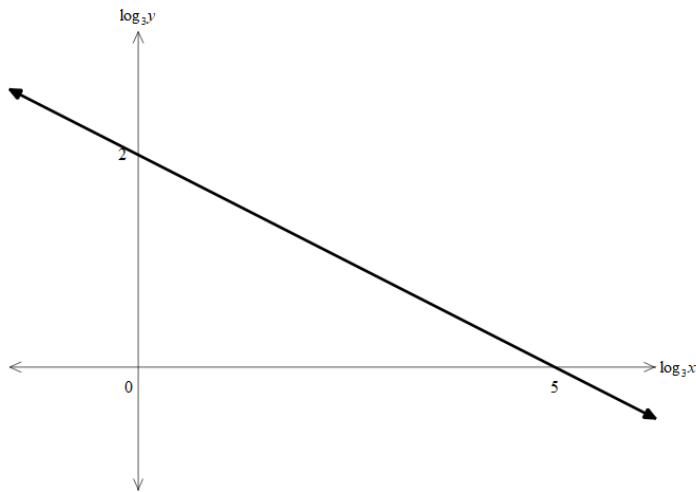
When was the particle at rest?

- A. $t = 0$
- B. $t = 1$ and $t = 4$
- C. $t = 2$ and $t = 4$
- D. $t = 1, t = 2$ and $t = 4$

7. The product of the 1st term and 2nd term of a geometric sequence is 48 while the product of the 3rd term and the 4th term of the sequence is 3888. What is the product of the 4th term and the 5th term of the sequence?

- A. 11664
- B. 15552
- C. 34992
- D. 186624

8. The graph shown below shows the linear relation between $\log_3 x$ and $\log_3 y$. Which of the following is true?

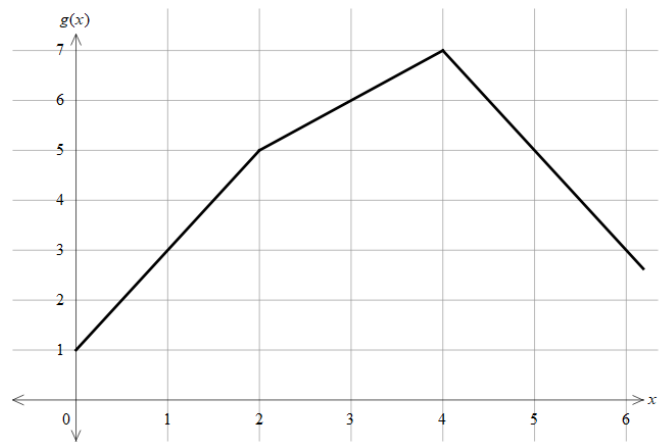
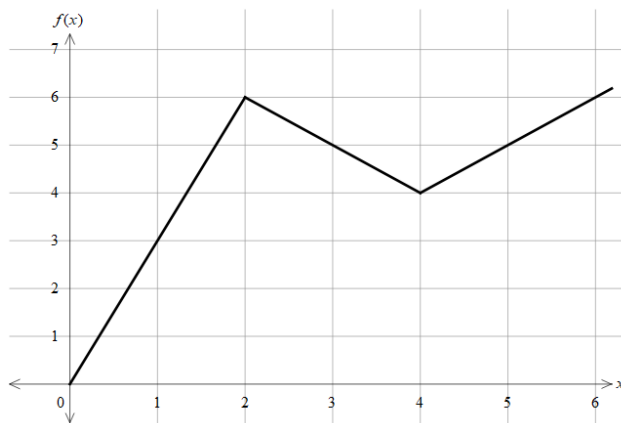


- A. $x^2 y^5 = 59049$
- B. $x^5 y^2 = 59049$
- C. $x^2 + y^5 = 59049$
- D. $x^5 + y^2 = 59049$

9. Find the probability that $-1 \leq 2 \sin \theta + 1 \leq 2$, where θ is chosen randomly in the range $0 \leq \theta \leq 2\pi$.

- A. $\frac{1}{3}$
 B. $\frac{1}{2}$
 C. $\frac{2}{3}$
 D. $\frac{4}{5}$

10. The graphs of the functions $f(x)$ and $g(x)$ are displayed below. Let $I(x) = f(g(x))$ and $J(x) = g(f(x))$. Which of the following has the highest value?



- A. $I'(1)$
 B. $I'(5)$
 C. $J'(1)$
 D. $J'(5)$

Trial Higher School Certificate Examination

Mathematics Advanced

Section II Answer Book

90 marks

Attempt Questions 11-35

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the space provided. These spaces provide guidance for the expected length of response.
- Your response should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet.

If you use this space, clearly indicate which question you are answering.

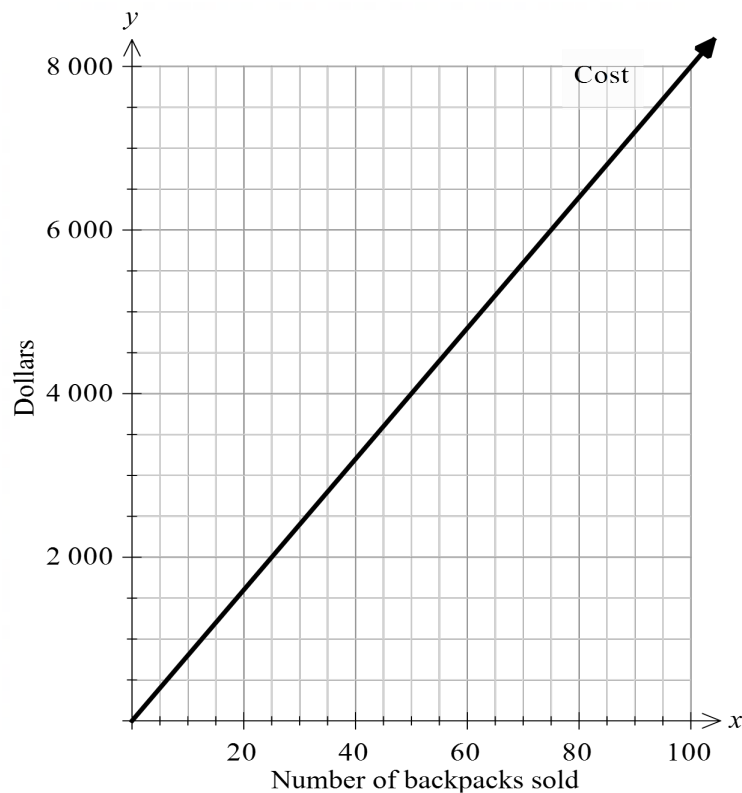
Question 11 (4 marks)

Jaimie owns a company producing and selling backpacks. The income function is $y = 80x$, where x is the number of backpacks sold. The cost of producing these backpacks includes a set-up cost of \$4500 and additional costs of \$30 per backpack.

- (a) Write the cost function in the form $y = mx + b$. 1

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- (b) Add the cost function to the set of axes below. 1



- (c) Hence, or otherwise, determine Jaimie's break-even point? 2

(i.e. how many backpacks does he need to sell to cover his costs?)

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Question 12 (2 marks)

Daniel has a bag that contains 5 red, and 8 black balls. Two balls are drawn randomly from the bag.

- (a) Find the probability that the first ball drawn is red. 1

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- (b) Find the probability that both balls drawn are the same colour. 1

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Question 13 (2 marks)

Let $f(x)$ be a function such that $\int_0^3 f(x) dx = 8$.

- (a) Find the value of $\int_0^3 2f(x) dx$. 1

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- (b) If $\int_c^d f(x-2) dx = 8$, find the values of c and d . 1

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Question 14 (2 marks)

Make y the subject of the formula for $k = \frac{3x - y}{y}$.

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Question 15 (2 marks)

Differentiate $5x^2 \tan\left(\frac{1}{x}\right)$.

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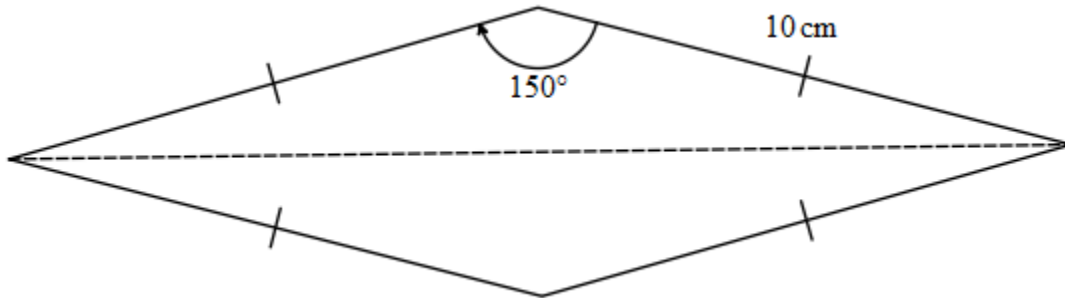
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Question 16 (2 marks)

A side of a rhombus is 10 cm and one of its angles is 150° as shown.

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Calculate its area.



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Question 17 (2 marks)

The table below shows the probability distribution of a discrete random variable X .

x	0	2	4	5	8	9
$P(X = x)$	k^2	0.16	0.18	0.3	k	0.12

- (a) Show that $k = 0.2$.

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- (b) Calculate $E(X)$.

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Question 18 (2 marks)

A sector has radius length of 20 cm and the angle subtended at the centre is 50° . **2**

The radius of this sector is increased by 25% and its angle at the centre is decreased by $k\%$. If the area of the sector remains unchanged find the value of k .

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Question 19 (2 marks)

(a) Show that $x^3 - 5x + \frac{xe^{-x^2}}{1+x^4}$ is an odd function. **1**

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(b) Hence, or otherwise, evaluate $\int_{-5}^5 x^3 - 5x + \frac{xe^{-x^2}}{1+x^4} dx$. **1**

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Question 20 (2 marks)

A researcher defined Scale A and Scale B to represent the magnitude of explosions as shown in the table below. **2**

Scale	Formula
A	$M = \log_4 E$
B	$N = \log_8 E$

It is given that E is the relative energy released by the explosion. If the magnitude of an explosion is 6.2 on Scale B, find the magnitude of the explosion on Scale A.

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Question 21 (3 marks)

The table below shows future value factors for an annuity of \$1.

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	A	B	C	D	E	F	G
1	Table of future value interest factors						
2	Periods\Rate	Interest rate per period					
3		0.50%	1.00%	1.50%	2.00%	3.00%	6.00%
4	5	5.0503	5.1010	5.1523	5.2040	5.3091	5.6371
5	10	10.2280	10.4622	10.7027	10.9497	11.4639	13.1808
6	20	20.9791	22.0190	23.1237	24.2974	26.8704	36.7856
7	40	44.1588	48.8864	54.2679	60.4020	75.4013	154.7620
8	60	69.7700	81.6697	96.2147	114.0515	163.0534	533.1282

Wayne and Vicki each undertake an annuity at 6% p.a. for a period of 5 years.

- Wayne invests \$3000 per quarter, compounded quarterly.
- Vicki invests \$1000 per month, compounded monthly.

Using supporting calculations, determine who will earn the greater amount of interest over the 5 years.

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Question 22 (3 marks)

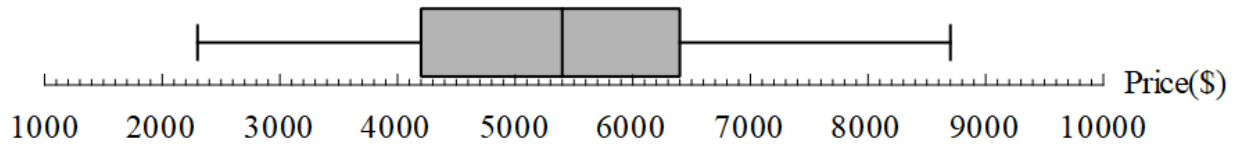
By factorising or otherwise, sketch the graph of $y = x^3 - 5x^2 - 4x + 20$ showing all intercepts with the axes. You do not need to find any stationary points. **3**

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Question 23 (5 marks)

There are 30 paintings in a warehouse. The box-and-whisker diagram below shows the prices of the paintings inside the warehouse.



- (a) Find the interquartile range. 1

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- (b) Show there are no outliers. 2

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- (c) It is given that the mean is \$5500 dollars. Four paintings of respective prices \$3400, \$3500, \$5700 and \$6800 are now donated to an art gallery. Find the mean and the median of the prices of the remaining paintings in the warehouse. 2

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Question 24 (4marks)

- (a) Prove that $(1 - \sin x)(\sec x + \tan x) \equiv \cos x$. **2**

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- (b) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \sin^2 x (1 - \sin x)(\sec x + \tan x) dx$. **2**

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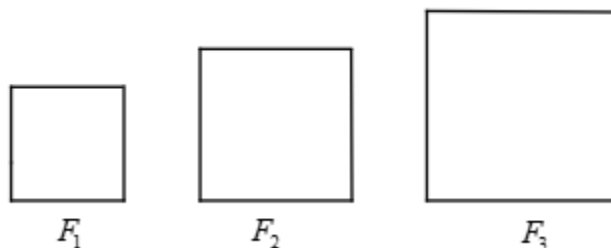
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Question 25 (3 marks)

In the diagram below, F_1, F_2, F_3 are square frames. The perimeter of the first frame is 16cm. The perimeter of each square frame afterwards is 4cm longer than the perimeter of the previous frame.



- (a) Find the perimeter of F_{12} . 1

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- (b) A thin metal wire of length 2000cm is cut into pieces and these pieces are then bent to form the above square frames. 2

Find the greatest number of distinct square frames that can be formed.

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Question 26 (3 marks)

Let X be a continuous random variable with probability density function

$$f(x) = \frac{1}{2}e^{-\frac{x}{2}} \text{ for } x > 0.$$

- (a) Find the cumulative distribution function of X . **2**

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- (b) Hence, or otherwise, find $P(X > 2)$ leaving your answer as a percentage to 2 decimal places. **1**

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Question 27 (3 marks)

A rational function $f(x)$ has the following properties,

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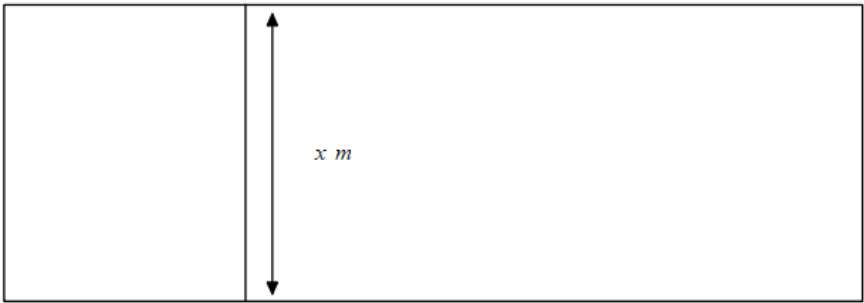
- The horizontal asymptote of its graph is $y = 0$
- The vertical asymptotes of its graph are $x = -2$ and $x = 2$
- The table below shows the first and second derivatives at various points.

	$x < -2$	$-2 < x < 0$	$x = 0$	$0 < x < 2$	$x > 2$
$f(x)$			1		
$f'(x)$	< 0	< 0	0	> 0	> 0
$f''(x)$	< 0	> 0	> 0	> 0	< 0 .

Sketch $y = f(x)$, using the properties in the table above.

Question 28 (4 marks)

A farmer has 156m of fencing to separate a paddock into two separate areas as shown in the diagram below.



- (a) Show that $A = 78x - \frac{3x^2}{2}$. 1

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- (b) Hence, or otherwise, find the maximum area of the paddock. 3

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Question 29 (5 marks)

The positive numbers $3, k, \frac{3}{2}$, form the start of a geometric sequence.

- (a) Show that $k = \frac{3}{\sqrt{2}}$. **1**

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- (b) Find the 7th term of this sequence. **1**

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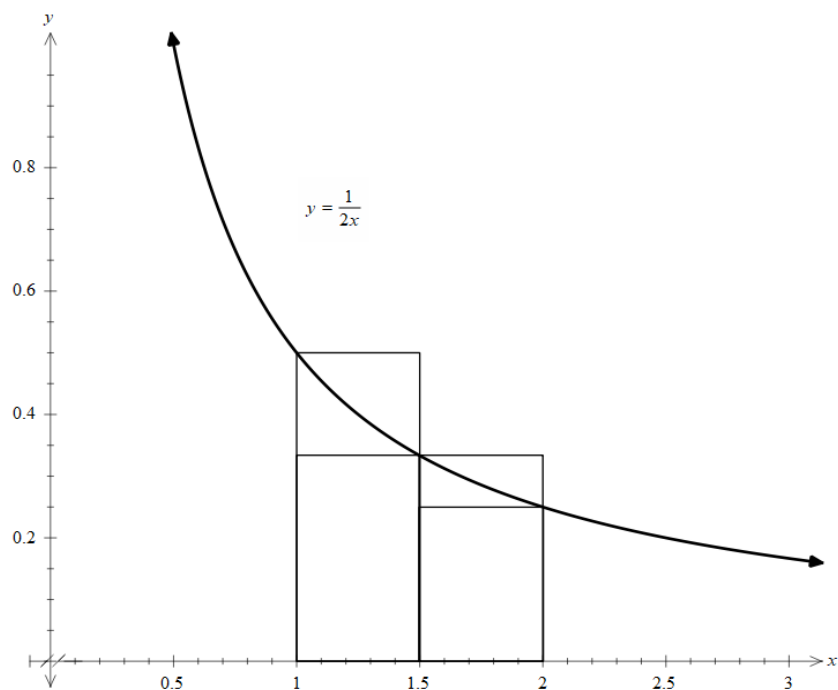
- (c) Find S_{∞} , expressing your answer in the form $p + \sqrt{q}$ where p and q are integers. **3**

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Question 30 (3 marks)

The graph of $f(x) = \frac{1}{2x}$ is shown below.

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By considering the areas of the rectangles, show that $\frac{7}{24} < \ln \sqrt{2} < \frac{5}{12}$.

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Question 31 (5 marks)

The price $P(t)$ in cents per litre of unleaded petrol during an average year in Broome WA, can be modelled by the function $P(t) = 180 + 44 \sin\left(\frac{2\pi t}{183}\right)$ where t is the number of days after 22 March 2023, for $0 \leq t \leq 366$.

- (a) What is the maximum price of petrol during the year? **1**

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- (b) Sketch the function $P(t)$ for $0 \leq t \leq 366$. **2**

Question 31 continues on page 26

Question 31 continued

- (c) What are the values of t for when petrol will cost 202 cents per litre.

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End of Question 31

Question 32 (6 marks)

Amanda has borrowed \$25 000 from a bank with interest accrued and repayments made monthly. The following table lists the progress for the first 6 months of the loan.

Month Number	Opening Balance	Monthly interest	Monthly Repayment	End Balance
1	\$25,000.00	\$112.50	\$400.00	\$24,712.50
2	\$24,712.50	\$111.21	\$400.00	\$24,423.71
3	\$24,423.71	\$109.91	\$400.00	\$24,133.61
4	\$24,133.61	\$108.60	\$400.00	\$23,842.21
5	\$23,842.21	\$107.29	\$400.00	\$23,549.50
6	\$23,549.50	\$105.97	\$400.00	\$23,255.48
7		A		B

- (a) What is the per annum rate of interest?

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- (b) Determine the values of A and B.

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Question 32 continues on page 28

Question 32 continued

(c) After how many months will Amanda have paid off the loan?

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End of Question 32

Question 33 (5 marks)

Waiting times for travellers of StarJet at an airport can be up to four hours. The associated probability density function is show below.



- (a) State the mode of $f(x)$. 1

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- (b) Find $f(x)$ that represents the above probability distribution function. 2

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Question 33 continues on page 30

Question 33 continued

- (c) Find the probability that a traveller will wait between 120 and 210 minutes. **2**

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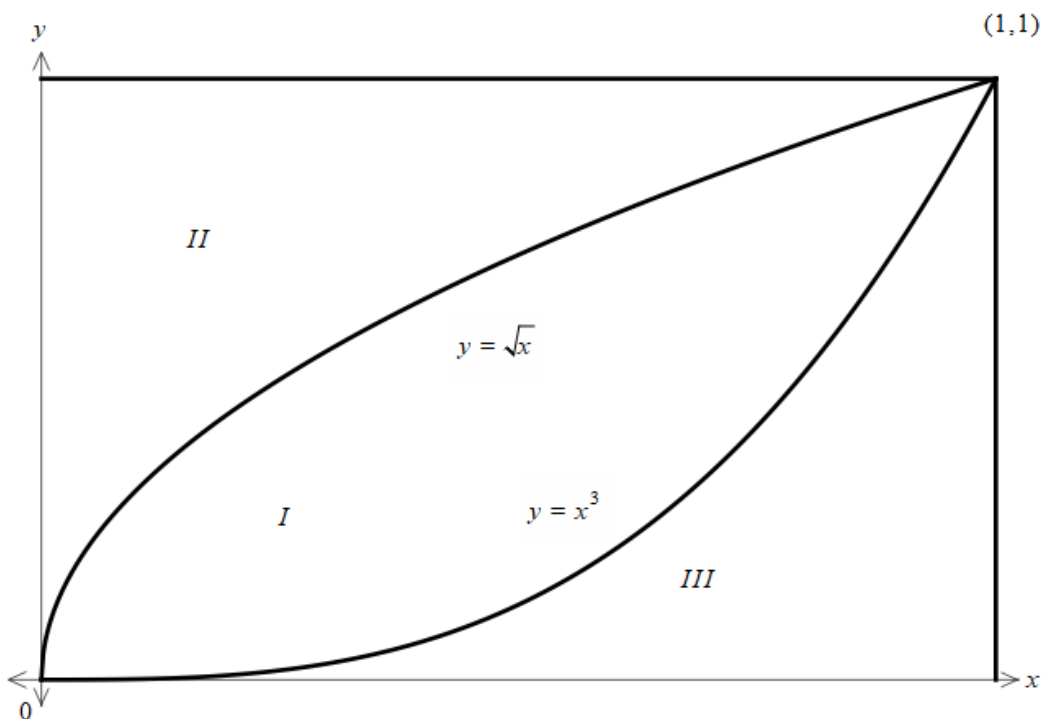
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End of Question 33

Question 34 (6 marks)

The diagram below shows a unit square target for shooting on the Cartesian Plane.

The target is divided into three regions, I, II and III by the curves $y = \sqrt{x}$ and $y = x^3$.



- (a) Find the areas of the three regions.

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Question 34 continues on page 32

Question 34 continued

- (b) The scores for hitting regions I, II and III are 5, 10 and 15 points respectively. **3**

Barry shoots randomly at the target twice and both shots hit the target.

Find the probability that if Barry scored 20 points that both his shots hit region II.

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End of Question 34

Question 35 (10 marks)

The population size P of a species of birds living in a wildlife preserve increases at a rate of

$$\frac{dP}{dt} = 9e^{\frac{t^2}{5}} - 2t \quad \text{for } t \geq 0,$$

where t is the time in months. It is known that the initial population of the birds is 34.

- (a) Use the trapezoidal rule with 4 sub intervals to estimate $\int_0^4 e^{\frac{t^2}{5}} dt$. **3**

Hence, show that $P = 218$ to the nearest integer at $t = 4$.

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Question 35 continues on page 34

A coal mine was recently built near the wildlife preserve and pollution from the mine affects the population of the birds from $t = 4$ onwards. An environmentalist models that the population size of the birds can then be approximated by $P = Ate^{-0.05t} - 100$, ($t \geq 4$).

- (b) Using part (a), show that $A = 97$. **1**

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- (c) Determine the maximum population size, leave your answer as an integer. **3**

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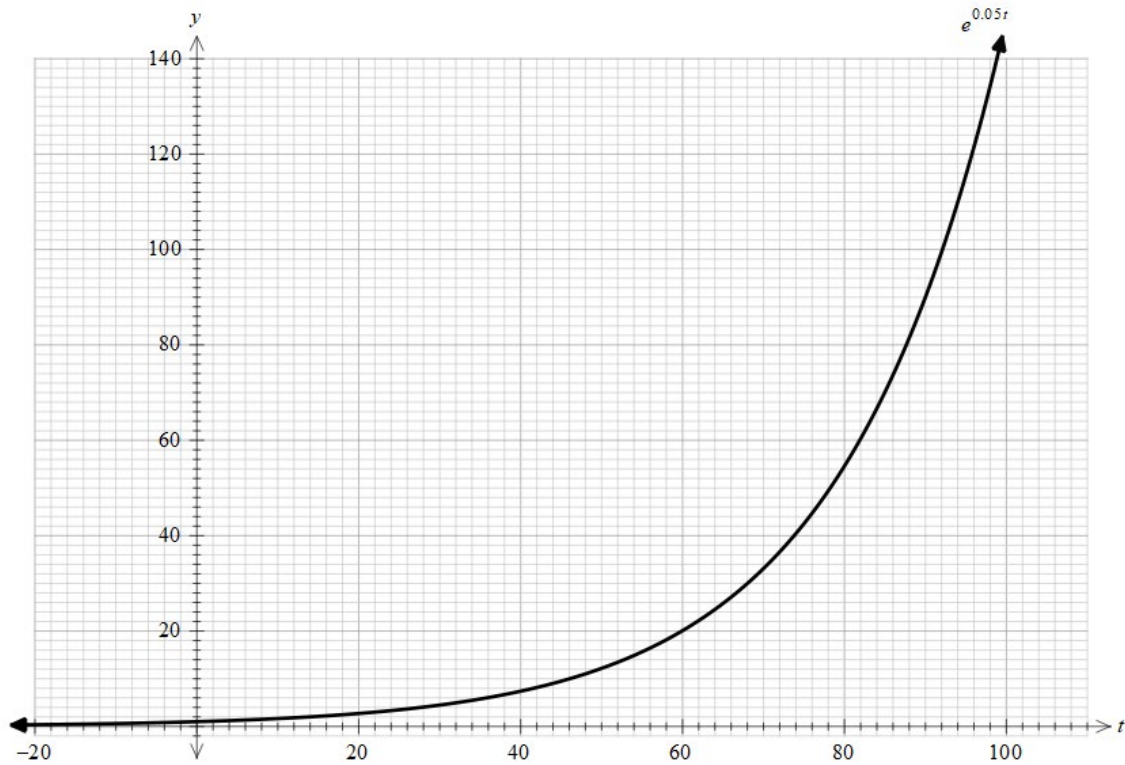
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Question 35 continues on page 35

Question 35 continued

- (d) Use the graph below to find the value of t , correct to the nearest whole number 3
for when the birds become extinct due to pollution.



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End of Exam

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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2023 Advanced Mathematics
TRIAL EXAMINATION
SUGGESTED SOLUTIONS

Section 1 Multiple Choice Answer Key

Question	Answer
1	D
2	B
3	A
4	B
5	B
6	C
7	C
8	A
9	C
10	C

Section II

Question 11(a)

Criteria	Marks
Provides correct answer	1

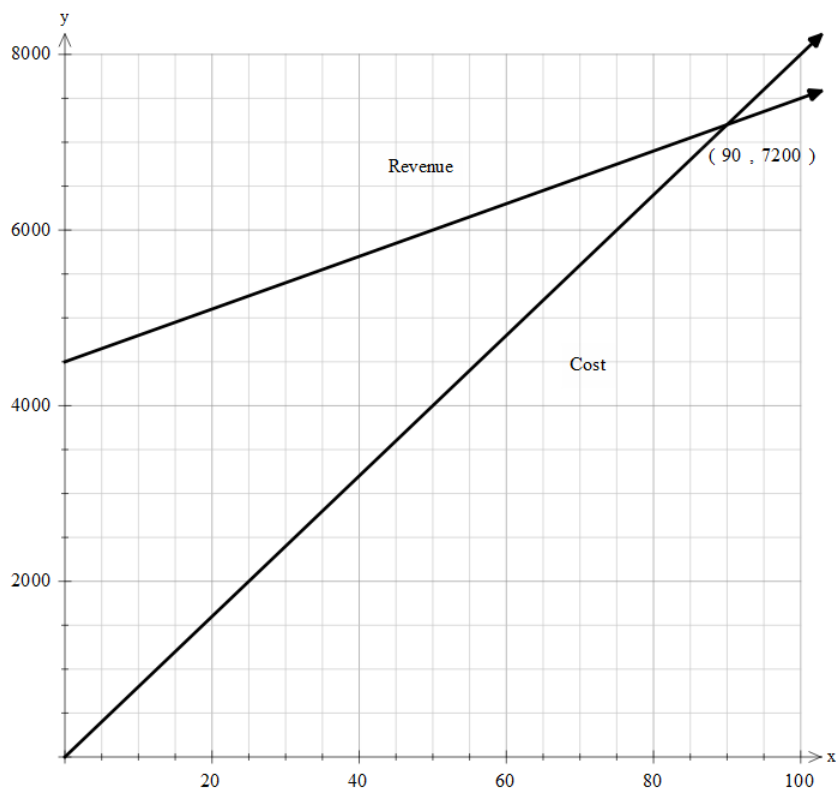
Sample Answer

$$y = 30x + 4500$$

Question 11 (b)

Criteria	Marks
Provides correct answer	1

Sample Answer



Question 11 (c)

Criteria	Marks
Provides correct solution	2
Attempts to solve $R(x) = C(x)$ or equivalent merit	1

Sample Answer

$$R(x) = C(x)$$

$$80x = 30x + 4500$$

$$50x = 4500$$

$$x = 90$$

i.e. Jaimie needs to sell 90 backpacks to breakeven

Question 12 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$P(R) = \frac{5}{13}$$

Question 12 (b)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$P(RR) \text{ or } P(BB)$$

$$= \frac{5}{13} \times \frac{4}{12} + \frac{8}{13} \times \frac{7}{12}$$

$$= \frac{19}{39}$$

Question 13 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$2 \times 8 = 16$$

Question 13 (b)

Criteria	Marks
Provides correct solution	1

Sample Answer

shift of 2 to the right

$$\therefore c = 2, d = 5$$

Question 14

Criteria	Marks
Provides correct solution	2
Uses one correct algebraic technique or equivalent merit	1

Sample Answer

$$\begin{aligned}
 ky &= 3x - y & k &= \frac{3x}{y} - 1 \\
 ky + y &= 3x \\
 y(k+1) &= 3x & \text{OR} & k+1 = \frac{3x}{y} \\
 y &= \frac{3x}{k+1} & \therefore y &= \frac{3x}{k+1}
 \end{aligned}$$

Question 15

Criteria	Marks
Provides correct solution	2
Attempts to use chain rule or differentiates $\tan\left(\frac{1}{x}\right)$ correctly	1

Sample Answer

$$u = 5x^2 \quad v = \tan\left(\frac{1}{x}\right)$$

$$u' = 10x \quad v' = -\frac{1}{x^2} \sec^2\left(\frac{1}{x}\right)$$

$$\therefore \frac{d}{dx}\left(5x^2 \tan\left(\frac{1}{x}\right)\right) = 10x \tan\left(\frac{1}{x}\right) - 5 \sec^2\left(\frac{1}{x}\right)$$

Question 16

Criteria	Marks
Provides correct solution	2
Attempts to use sine rule or equivalent merit	1

Sample Answer

$$A = 2 \times \frac{1}{2} ab \sin C$$

$$A = 10 \times 10 \times \sin(150^\circ)$$

$$A = 50 \text{ cm}^2$$

Question 17 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$k^2 + 0.16 + 0.18 + 0.3 + k + 0.12 = 1$$

$$k^2 + k + 0.76 = 1$$

$$k^2 + k - 0.24 = 0$$

$$(k - 0.2)(k + 1.2) = 0$$

$$\therefore k = 0.2 \text{ as } k > 0$$

Question 17 (b)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$E(X) = 0 \times 0.2^2 + 2 \times 0.16 + 4 \times 0.18 + 5 \times 0.3 + 8 \times 0.2 + 9 \times 0.12$$

$$E(X) = 5.22$$

Question 18

Criteria	Marks
Provides correct solution	2
Progress towards solution	1

Sample Answer

$$\frac{1}{2} \times 20^2 \times \frac{50}{360} = \frac{1}{2} \times 25^2 \times \frac{\theta}{360}$$

$$20000 = 625\theta$$

$$\theta = 32$$

$$\therefore \% \text{Difference} = \frac{18}{50} \times 100 = 36\%$$

$$\therefore k = 36$$

Question 19 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$f(-x) = (-x)^3 - 5(-x) + \frac{(-x)e^{-(-x)^2}}{1+(-x)^4}$$

$$f(-x) = -x^3 + 5x - \frac{xe^{-x^2}}{1+x^4}$$

$$f(-x) = -\left(x^3 - 5x + \frac{xe^{-x^2}}{1+x^4}\right)$$

$$\therefore f(-x) = -f(x)$$

$\therefore$ function is odd

Question 19 (b)

Criteria	Marks
Provides correct solution	1

Sample Answer

integrating an odd function over a balanced interval

$$\therefore \int_{-5}^5 x^3 - 5x + \frac{xe^{-x^2}}{1+x^4} dx = 0.$$

Question 20

Criteria	Marks
Provides correct solution	2
Progress towards solution	1

Sample Answer

$$6.2 = \log_8(E)$$

$$E = 8^{6.2}$$

$$M = \log_4(8^{6.2})$$

$$M = 9.3$$

Question 21

Criteria	Marks
Provides correct solution	3
Calculates one future value using the interest table, or equivalent merit	2
Finds one correct table value or equivalent merit	1

Sample Answer

Wayne: 1.5% per quarter, 20 periods

$$\therefore \$3000 \times 23.1237 = \$69371.10$$

$$\text{Interest earned} = 69371.1 - 3000 \times 20$$

$$= \$9371.10$$

Vicki: 0.5% per month, 60 periods

$$\therefore \$1000 \times 69.7700 = \$69770$$

$$\text{Interest earned} = 69770 - 1000 \times 60$$

$$= \$9770$$

$\therefore$ Vicki earned more interest

Question 22

Criteria	Marks
Provides correct solution	3
Provides correct x and y intercepts or correct shape, or equivalent merit	2
Factorises correctly or finds correct y intercept	1

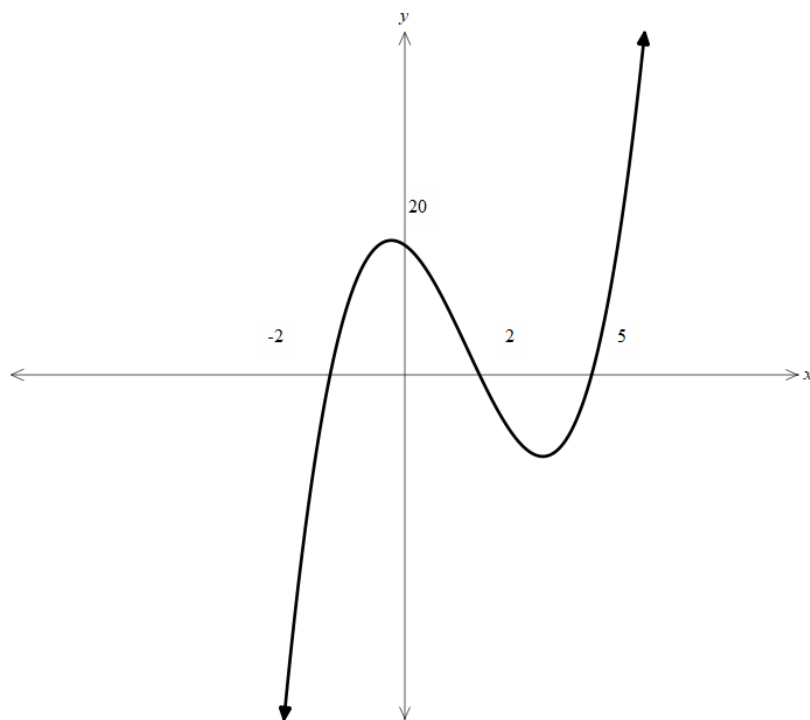
Sample Answer

$$y = x^3 - 5x^2 - 4x + 20$$

$$y = x^2(x-5) - 4(x-5)$$

$$y = (x^2 - 4)(x-5)$$

$$y = (x-2)(x+2)(x-5)$$



Note: Location of maximum turning point does not matter

Question 23 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$Q_3 = 6400$$

$$Q_1 = 4200$$

$$IQR = 6400 - 4200$$

$$IQR = 2200$$

Question 23 (b)

Criteria	Marks
Provides correct solution	2
Attempts to use IQR formula or equivalent merit	1

Sample Answer

$$\text{min} = 2300$$

$$\text{max} = 8700$$

$$Q_1 - 1.5 \times IQR = 4200 - 1.5 \times 2200$$

$$= 900 < 2300$$

$$Q_3 + 1.5 \times IQR = 6400 + 1.5 \times 2200$$

$$= 9700 > 8700$$

$\therefore$ no outliers

Question 23 (c)

Criteria	Marks
• Provides correct solution	2
• Finds correct mean or median	1

Sample Answer

median = 5400

2 paintings below and 2 paintings above the median are removed

∴ median remains unchanged

i.e. median = \$5400

total sum of prices = 5500×30

= 165000

∴ new total = $165000 - 3400 - 3500 - 5700 - 6800$

= 145600

new mean = $\frac{145600}{26}$

= \$5600

Question 24 (a)

Criteria	Marks
• Provides correct solution	2
• Progress towards solution	1

Sample Answer

$$LHS : (1 - \sin x) \left(\frac{1}{\cos x} + \frac{\sin x}{\cos x} \right)$$

$$\equiv (1 - \sin x) \left(\frac{1 + \sin x}{\cos x} \right)$$

$$\equiv \left(\frac{1 - \sin^2 x}{\cos x} \right)$$

$$\equiv \left(\frac{\cos^2 x}{\cos x} \right)$$

$$\equiv \cos x$$

$$= RHS$$

Question 24 (b)

Criteria	Marks
Provides correct solution	2
Provides correct integral or equivalent merit	1

Sample Answer

from (a)

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^2 x (1 - \sin x) (\sec x + \tan x) dx &= \int_0^{\frac{\pi}{2}} \sin^2 x \cos x dx \\&= \frac{\sin^3 x}{3} \Bigg|_0^{\frac{\pi}{2}} \\&= \frac{1}{3} \left(\sin^3 \left(\frac{\pi}{2} \right) - \sin^3 (0) \right) \\&= \frac{1}{3} (1 - 0) \\&= \frac{1}{3}\end{aligned}$$

Question 25 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$a = 16, d = 4, n = 12$$

$$T_{12} = 16 + (12 - 1)4$$

$$T_{12} = 60 \text{ cm}$$

Question 25 (b)

Criteria	Marks
Provides correct solution	2
Forms a quadratic equation or equivalent merit	1

Sample Answer

$$S_n = 2000, a = 16, d = 4$$

$$2000 = \frac{n}{2}(2(16) + 4(n - 1))$$

$$4000 = n(28 + 4n)$$

$$4n^2 + 28n - 4000 = 0$$

$$n = \frac{-28 \pm \sqrt{28^2 - 4(4)(-4000)}}{2(4)}$$

$$n \approx -35.31587654\dots, 28.31587654\dots$$

$$n \in \mathbb{N}^+$$

$$\therefore n = 28$$

i.e. 28 distinct square frames can be formed

Question 26 (a)

Criteria	Marks
• Provides correct solution	2
• Progress towards solution	1

Sample Answer

$$\begin{aligned}
 & \int_0^X \frac{1}{2} e^{-\frac{x}{2}} dx \\
 &= -e^{-\frac{x}{2}} \Big|_{x=0}^{x=X} \\
 &= -e^{-\frac{X}{2}} - (-e^0) \\
 &= 1 - e^{-\frac{X}{2}} \\
 \therefore CDF &= \begin{cases} 1 - e^{-\frac{x}{2}}, & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}
 \end{aligned}$$

Question 26 (b)

Criteria	Marks
• Provides correct solution	1

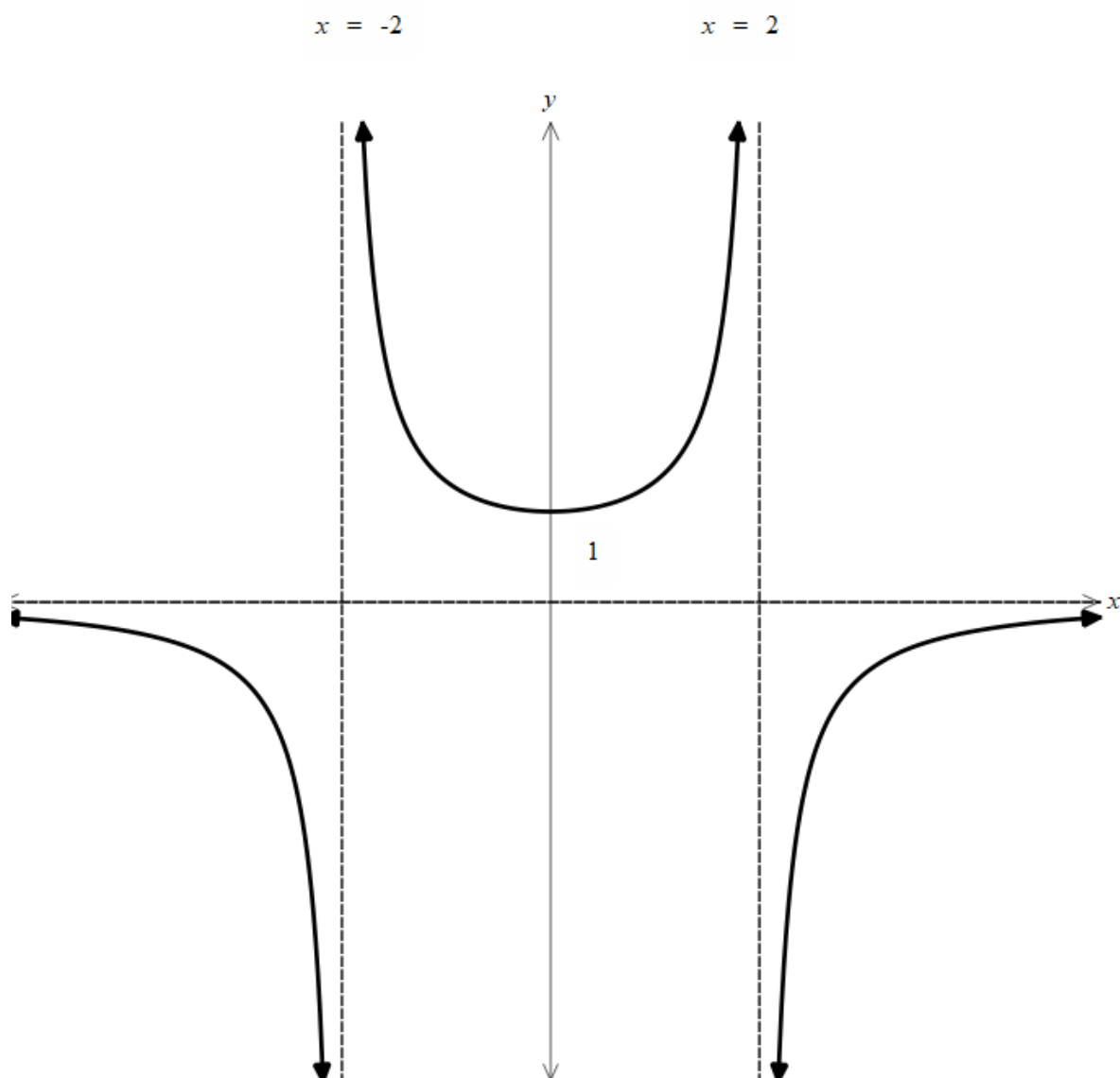
Sample Answer

$$\begin{aligned}
 P(X > 2) &= 1 - P(X \leq 2) \\
 &= 1 - (1 - e^{-1}) \\
 &\approx 0.3678794412 \\
 &= 36.79\% \text{ (2 d.p.)}
 \end{aligned}$$

Question 27

Criteria	Marks
Provides correct solution, showing all essential features	3
Shows asymptotes on graph and 3 correct points from table	2
Shows asymptotes on graph or 1 correct point from table	1

Sample Answer



Question 28 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$P = 3x + 2y$$

$$156 = 3x + 2y$$

$$y = 78 - \frac{3x}{2}$$

$$A = xy$$

$$A = x \left(78 - \frac{3x}{2} \right)$$

$$A = 78x - \frac{3x^2}{2}$$

Question 28 (b)

Criteria	Marks
Provides correct solution	3
Finds correct stationary value and justifies it is a maximum	2
Differentiates correctly and sets $\frac{dA}{dx} = 0$, or equivalent merit	1

Sample Answer

$$\frac{dA}{dx} = 78 - 3x$$

$$\frac{dA}{dx} = 0$$

$$\therefore 78 - 3x = 0$$

$$3x = 78$$

$$x = 26$$

$$\frac{d^2 A}{dx^2} = -3$$

$$\frac{d^2 A}{dx^2}(26) < 0$$

$\therefore$ concave down $\rightarrow$ maximum

$$\therefore \text{maximum Area} = 78(26) - \frac{3(26)^2}{2}$$

$$= 1014 \text{ m}^2$$

maximum occurs at the vertex

as function is concave down parabola

$$x = -\frac{b}{2a}$$

OR

$$x = -\frac{78}{2(-\frac{3}{2})}$$

$$x = 26$$

$$\therefore A(26) = 1014 \text{ m}^2$$

Question 29 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$\begin{aligned}
 3r^2 &= \frac{3}{2} \\
 r^2 &= \frac{1}{2} \\
 r &= \frac{1}{\sqrt{2}} \text{ as } r > 0 \\
 \therefore k &= \frac{3}{\sqrt{2}}
 \end{aligned}
 \qquad
 \text{OR}
 \qquad
 \begin{aligned}
 \frac{k}{3} &= \frac{3/2}{k} \\
 k^2 &= \frac{9}{2} \\
 k &= \frac{3}{\sqrt{2}}, \text{ as } k > 0
 \end{aligned}$$

Question 29 (b)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$\begin{aligned}
 a &= 3, r = \frac{1}{\sqrt{2}}, n = 7 \\
 T_7 &= 3 \left(\frac{1}{\sqrt{2}} \right)^6 \\
 T_7 &= \frac{3}{8}
 \end{aligned}$$

Question 29 (c)

Criteria	Marks
• Provides correct solution	3
• Progress towards solution	2
• Attempts to use limiting sum formula or equivalent merit	1

Sample Answer

$$a = 3, r = \frac{1}{\sqrt{2}}$$

$$S_{\infty} = \frac{3}{1 - \frac{1}{\sqrt{2}}}$$

$$S_{\infty} = \frac{3\sqrt{2}}{\sqrt{2} - 1}$$

$$S_{\infty} = \frac{3\sqrt{2}}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$$

$$S_{\infty} = \frac{6 + 3\sqrt{2}}{2 - 1}$$

$$S_{\infty} = 6 + \sqrt{18}$$

$$\therefore p = 6, q = 18$$

Question 30

Criteria	Marks
• Provides correct solution	3
• Progress towards solution	2
• Sets up correct inequality or calculates one correct value.	1

Sample Answer

$$\text{lower bound} < \int_1^2 \frac{1}{2x} dx < \text{upper bound}$$

$$\therefore \frac{1}{3} \times \frac{1}{2} + \frac{1}{4} \times \frac{1}{2} < \int_1^2 \frac{1}{2x} dx < \frac{1}{2} \times \frac{1}{2} + \frac{1}{3} \times \frac{1}{2}$$

$$\therefore \frac{7}{24} < \frac{1}{2} \ln x \Big|_1^2 < \frac{5}{12}$$

$$\frac{7}{24} < \frac{1}{2} (\ln 2 - \ln 1) < \frac{5}{12}$$

$$\frac{7}{24} < \frac{1}{2} \ln 2 < \frac{5}{12}$$

$$\frac{7}{24} < \ln \sqrt{2} < \frac{5}{12}$$

Question 31 (a)

Criteria	Marks
Shows correct progress	1

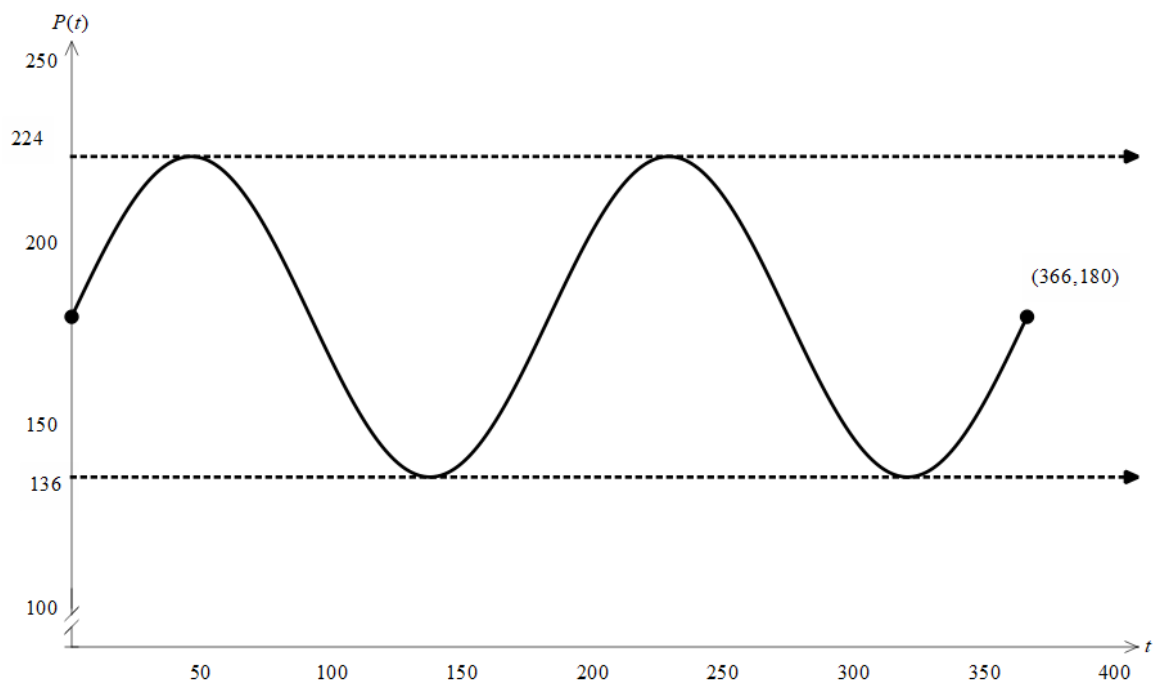
Sample Answer

$$\text{max} = 180 + 44$$

$$\text{max} = 224$$

Question 31 (b)

Criteria	Marks
Provides correct solution	2
Provides a sketch showing some correct features	1

Sample Answer

Question 31 (c)

Criteria	Marks
• Provides correct solution	2
• Progress towards solution	1

Sample Answer

$$202 = 180 + 44 \sin\left(\frac{2\pi t}{183}\right)$$

$$22 = 44 \sin\left(\frac{2\pi t}{183}\right)$$

$$\frac{1}{2} = \sin\left(\frac{2\pi t}{183}\right)$$

$$\frac{2\pi t}{183} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$$

$$t = 15.25, 76.25, 198.25, 259.25$$

i.e. during the 15th, 76th, 198th and 259th day

Question 32 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$25112.50 = 25000(1+r)$$

$$1+r = \frac{25112.50}{25000}$$

$$r = \frac{25112.50}{25000} - 1$$

$$r = 0.45\% \text{ per month}$$

$$r = 5.4\% \text{ per annum.}$$

Question 32 (b)

Criteria	Marks
Provides correct solution	2
Finds correct value of A or B	1

Sample Answer

$$A = 23255.48(0.0045)$$

$$A = 104.64966$$

$$A = \$104.65(2d.p.)$$

$$B = 23255.48 + 104.65 - 400$$

$$B = \$22960.13$$

Question 32 (c)

Criteria	Marks
Provides correct solution	3
Writes the equation in simplest exponential form	2
Finds an expression for A_n or equivalent merit	1

Sample Answer

$$A_1 = 25000(1.0045) - 400$$

$$A_2 = A_1(1.0045) - 400$$

$$A_2 = 25000(1.0045)^2 - 400(1.0045) - 400$$

$$\vdots$$

$$A_n = 25000(1.0045)^n - 400(1.0045)^{n-1} - 400(1.0045)^{n-2} - \dots - 400$$

$$A_n = 25000(1.0045)^n - 400(1 + 1.0045 + \dots + 1.0045^{n-2} + 1.0045^{n-1})$$

$$0 = 25000(1.0045)^n - 400 \left[\frac{1(1.0045^n - 1)}{1.0045 - 1} \right]$$

$$0 = 112.5(1.0045)^n - 400(1.0045)^n + 400$$

$$287.5(1.0045)^n = 400$$

$$1.0045^n = \frac{400}{287.5}$$

$$n = \log_{1.0045} \left(\frac{400}{287.5} \right)$$

$$n \approx 73.55203881\dots$$

$$\therefore n = 74$$

i.e. will take 74 months to pay off the loan

Question 33 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

Mode is 3

Question 33 (b)

Criteria	Marks
Provides correct solution	2
Provides one correct function	1

Sample Answer

$$f(x) = \begin{cases} \frac{x}{6}, & 0 \leq x \leq 3 \\ 2 - \frac{x}{2}, & 3 < x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

Question 33 (c)

Criteria	Marks
• Provides correct solution	2
• Sets up correct integral or equivalent merit	1

Sample Answer

$$P(120 < X < 210) = P(2 < X < 3.5)$$

$$= \int_2^3 \frac{x}{6} dx + \int_3^{3.5} 2 - \frac{x}{2} dx$$

$$= \frac{x^2}{12} \Big|_2^3 + \left(2x - \frac{x^2}{4} \right) \Big|_3^{3.5}$$

$$= \left(\frac{9}{12} - \frac{4}{12} \right) + \left(7 - \frac{12.25}{4} - \left(6 - \frac{9}{4} \right) \right) \quad \text{OR}$$

$$= \frac{5}{12} + \frac{3}{16}$$

$$= \frac{29}{48}$$

Area under the curve (both are trapezoids)

$$A = \frac{1}{2} \times 1 \left(\frac{1}{3} + \frac{1}{2} \right) + \frac{1}{2} \times \frac{1}{2} \left(\frac{1}{2} + \frac{1}{4} \right)$$

$$A = \frac{5}{12} + \frac{3}{16}$$

$$A = \frac{29}{48}$$

Question 34 (a)

Criteria	Marks
• Provides correct solution	3
• Finds two correct areas	2
• Finds one correct area	1

Sample Answer

$$Area(III) = \int_0^1 x^3 \, dx$$

$$= \frac{x^4}{4} \Big|_0^1$$

$$= \frac{1}{4}$$

$$Area(I) = \int_0^1 \sqrt{x} - x^3 \, dx$$

$$= \frac{2}{3} \sqrt{x^3} - \frac{x^4}{4} \Big|_0^1$$

$$= \frac{5}{12}$$

$$Area(II) = 1 - \frac{1}{4} - \frac{5}{12}$$

$$= \frac{1}{3}$$

Question 34 (b)

Criteria	Marks
• Provides correct solution	3
• Progress towards solution	2
• Finds a correct probability for 20 points or attempts to use conditional probability	1

Sample Answer

$$P(20 \text{ points}) = P(I \text{ \& } III) \text{ or } P(III \text{ \& } I) \text{ or } P(II \text{ \& } II)$$

$$= \frac{5}{12} \times \frac{1}{4} + \frac{1}{4} \times \frac{5}{12} + \frac{1}{3} \times \frac{1}{3}$$

$$= \frac{23}{72}$$

$$P(\text{region II twice} \cap 20 \text{ points}) = \frac{1}{9}$$

$$\therefore P(\text{region II twice} | 20 \text{ points}) = \frac{\frac{1}{9}}{\frac{23}{72}}$$

$$= \frac{8}{23}$$

Question 35 (a)

Criteria	Marks
• Provides correct solution	3
• Provides correct anti-derivative or equivalent merit	2
• Correctly uses trapezoidal rule	1

Sample Answer

x	0	1	2	3	4
$f(x)$	1	1.2214...	2.2255..	6.0496...	24.5325..

$$\therefore \int_0^4 e^{\frac{t^2}{5}} dt \approx \frac{4-0}{2(4)} (1 + 24.5325 + 2(1.2214 + 2.2255 + 6.0496))$$

$$\therefore \int_0^4 e^{\frac{t^2}{5}} dt \approx 22.26275$$

$$\therefore P|_{t=4} - P|_{t=0} = \int_0^4 9e^{\frac{t^2}{5}} - 2t dt$$

$$\therefore P|_{t=4} = \int_0^4 9e^{\frac{t^2}{5}} - 2t dt + 34$$

$$\therefore P|_{t=4} = 9 \int_0^4 e^{\frac{t^2}{5}} dt - t^2 \Big|_0^4 + 34$$

$$\therefore P|_{t=4} = 9 \times 22.26275 - 16 + 34$$

$$\therefore P|_{t=4} \approx 218.36475$$

$$\therefore P = 218, \text{ when } t = 4$$

Question 35 (b)

Criteria	Marks
• Provides correct solution	1

Sample Answer

$$218 = 4Ae^{-0.2} - 100$$

$$318 = 4Ae^{-0.2}$$

$$A = \frac{318}{4e^{-0.2}}$$

$$A \approx 97.10151927..$$

$$A = 97$$

Question 35 (c)

Criteria	Marks
Provides correct solution	3
Finds correct t value of the stationary point	2
Finds $P'(t)$ correctly	1

Sample Answer

$$P(t) = 97te^{-0.05t} - 100$$

$$P'(t) = 97e^{-0.05t} - \frac{97}{20}te^{-0.05t}$$

$$P'(t) = 0$$

$$\therefore 97e^{-0.05t} \left(1 - \frac{t}{20}\right) = 0$$

$$e^{-0.05t} \neq 0$$

$$\therefore t = 20$$

t	19	20	21
$P'(t)$	1.87569...	0	-1.6979...
	increasing	Stationary	decreasing

$\therefore t = 20$ is a maximum

$$P(20) = 97(20)e^{-1} - 100$$

$$P(20) = 613.6861159...$$

$\therefore \text{max population} = 613$

Question 35 (d)

Criteria	Marks
Provides correct solution	3
Progress towards solution	2
Substitutes $y = e^{0.05t}$ into P	1

Sample Answer

Using $y = e^{0.05t}$

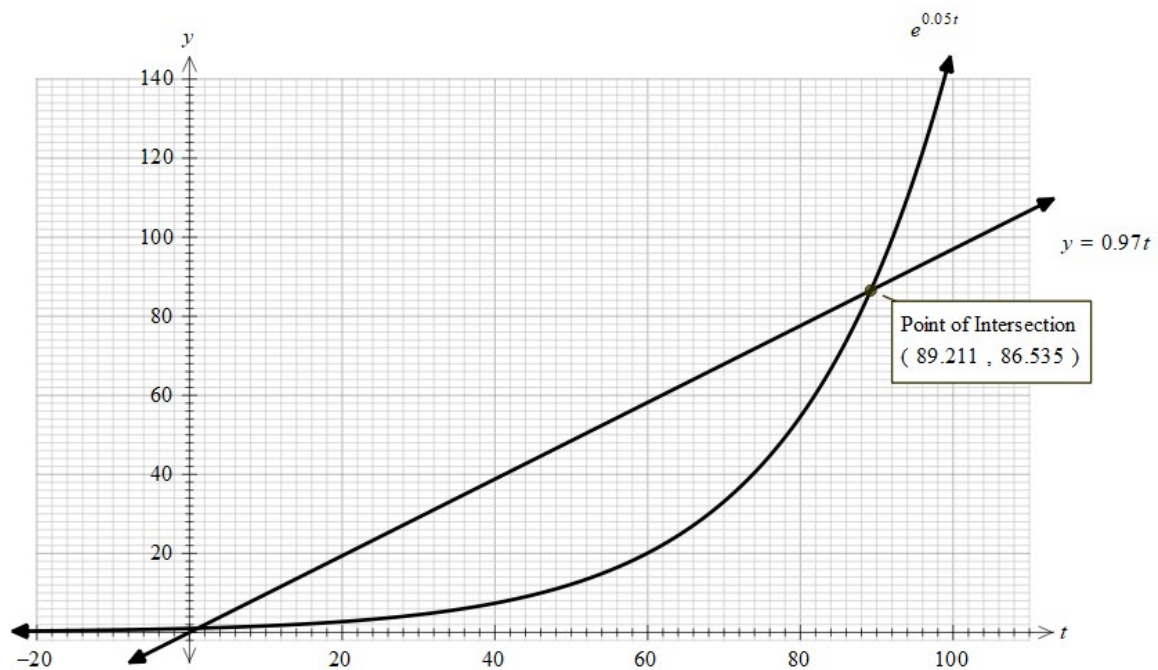
$$\therefore \frac{1}{y} = e^{-0.05t}$$

Substitute into $97te^{-0.05t} - 100 = 0$

$$\therefore \frac{97t}{y} = 100$$

$$y = 0.97t.$$

The graphs of $y = e^{0.05t}$ and $y = 0.97t$ intersect at $t \approx 1$ and $t \approx 89$



as $t \geq 4$, the birds will become extinct when $97te^{-0.05t} - 100 = 0$ when $t \approx 89$

Note: Due to accuracy accept $88 < t < 90$

2023 PEM Trial HSC Advanced Mathematics

Mapping Grid

Section 1

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
1	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	2
2	1	MA-F1 Working with Functions	MA11-1	3
3	1	MA-F1 Working with Functions	MA11-1	2-3
4	1	MA-M1 Modelling Financial Situations	MA12-2	3
5	1	MA-S3 Random Variables	MA12-8	3-4
6	1	MA-E1 Logarithms and Exponentials	MA12-6	4
7	1	MA-M1 Modelling Financial Situations	MA12-4	4
8	1	MA-F1 Working with Functions MA-E1 Logarithms and Exponentials	MA11-2 MA11-6	4-6
9	1	MA-S1 Probability and Discrete Probability Distributions MA-T3 Trigonometric Functions and Graphs	MA11-7 MA12-5	5-6
10	1	MA-F1 Working with Functions MA-C1 Introduction to Differentiation	MA11-5	6

Section II

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
11 (a)	1	MA-F1 Working with Functions	MA11-2	2
11 (b)	1	MA-F1 Working with Functions	MA11-2	2-3
11 (c)	2	MA-F1 Working with Functions	MA11-2	3-4
12 (a)	1	MA-S1 Probability and Discrete Probability Distributions	MA11-7	2
12 (b)	1	MA-S1 Probability and Discrete Probability Distributions	MA11-7	2-3
13 (a)	1	MA-C4 Integral Calculus	MA12-7	2-3

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
13 (b)	1	MA-C4 Integral Calculus	MA12-7	3-4
14	2	MA-F1 Working with Functions	MA11-1	3-4
15	2	MA-C2 Differential Calculus	MA12-6	4
16	2	MA-T1 Trigonometry and Measure of Angles	MA11-3	3
17 (a)	1	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3-4
17 (b)	1	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3-4
18	2	MA-T1 Trigonometry and Measure of Angles	MA11-3	3-4
19 (a)	1	MA-F1 Working with Functions	MA11-1	3-4
19 (b)	1	MA-C4 Integral Calculus	MA12-7	4-5
20	2	MA-E1 Logarithms and Exponentials	MA11-6	3-4
21	3	MA-M1 Modelling Financial Situations	MA12-2	3-4
22	3	MA-F1 Working with Functions	MA11-1	4-5
23 (a)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	3
23 (b)	2	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	3-4
23 (c)	2	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	3-5
24 (a)	2	MA-T2 Trigonometric Functions and Identities	MA11-4	4
24 (b)	2	MA-C4 Integral Calculus	MA12-7	4-5
25 (a)	1	MA-M1 Modelling Financial Situations	MA12-4	4
25 (b)	2	MA-M1 Modelling Financial Situations	MA12-4	4-5
26 (a)	2	MA-S3 Random Variables	MA12-8	4-5
26 (b)	1	MA-S3 Random Variables	MA12-8	4-5
27	3	MA-F2 Graphing Techniques	MA12-1	5-6
28 (a)	1	MA-F1 Working with Functions	MA11-2	3-4
28 (b)	3	MA-C3 Applications of Differentiation	MA12-6	5-6
29 (a)	1	MA-M1 Modelling Financial Situations	MA12-4	3
29 (b)	1	MA-M1 Modelling Financial Situations	MA12-4	3
29 (c)	3	MA-M1 Modelling Financial Situations	MA12-4	4-5
30	3	MA-C4 Integral Calculus	MA12-7	4-6
31 (a)	1	MA-T3 Trigonometric Functions and Graphs	MA12-5	3
31 (b)	2	MA-T3 Trigonometric Functions and Graphs	MA12-5	4-5
31 (c)	2	MA-T3 Trigonometric Functions and Graphs	MA12-5	4-5
32 (a)	1	MA-M1 Modelling Financial Situations	MA12-2	3-4

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
32 (b)	2	MA-M1 Modelling Financial Situations	MA12-2	3-4
32 (c)	3	MA-M1 Modelling Financial Situations	MA12-2	5-6
33 (a)	1	MA-S3 Random Variables	MA12-8	2-3
33 (b)	2	MA-F1 Working with Functions	MA11-2	4
33 (c)	2	MA-S3 Random Variables	MA12-8	4-5
34 (a)	3	MA-C4 Integral Calculus	MA12-7	4-6
34 (b)	3	MA-S1 Probability and Discrete Probability Distributions	MA11-7	4-6
35 (a)	3	MA-C4 Integral Calculus	MA12-7	6
35 (b)	1	MA-E1 Logarithms and Exponentials	MA11-6	4
35 (c)	3	MA-C3 Applications of Differentiation	MA12-6	5-6
35 (d)	3	MA-F2 Graphing Techniques	MA12-1	6